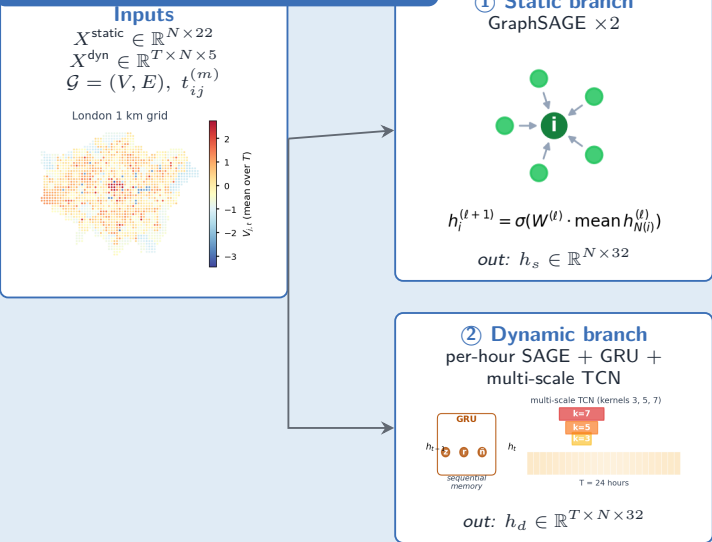
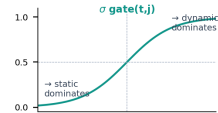


Stage A · Encoder (33,026 weights)



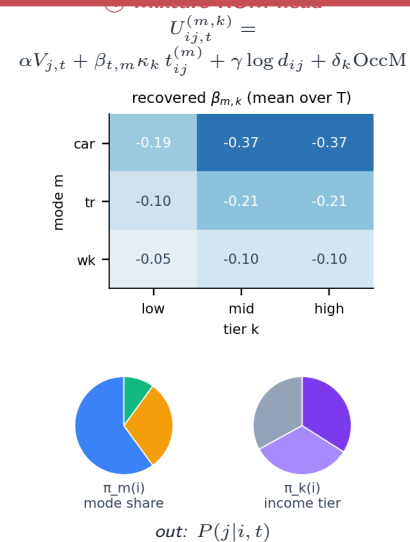
③ Gated fusion

σ gate per (t, j)



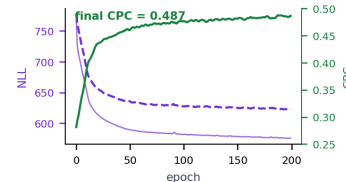
out: $V_{j,t} \in \mathbb{R}^{T \times N}$

Stage B · Behavioural choice + loss (77 interp. params)



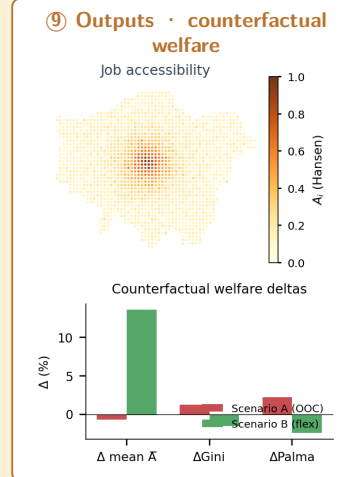
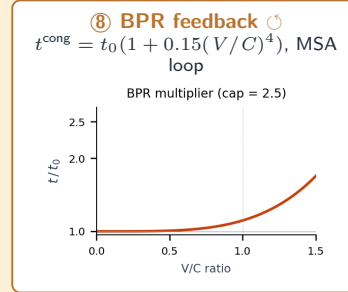
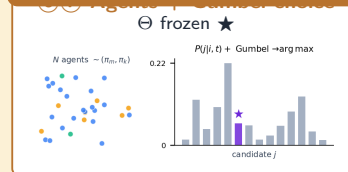
⑤ Cross-entropy loss

$$\mathcal{L} = - \sum_{i,j,t} F_{i,j,t} \log P(j|i, t)$$



train CPC = 0.488 (London 2019)

Stage C · Deployment + counterfactual (Θ frozen)



Trainable parameters: see Table 1. Utility + BPR equations: see §2.5–§2.6. Stage A learns deep ML weights via $\nabla_{\Theta} \mathcal{L}$; Stage B identifies 77 interpretable behavioural primitives; Stage C deploys frozen Θ with BPR closure for counterfactual analysis.